Basics of Nutrition for Sports Performance and Recovery: A Short Review

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ABSTRACT

There are many factors contributing to better sports performance including optimal training and periodization, enough sleep, proper rest, hydration. Sports nutrition plays a critical role in improving sports performance with injury/illness minimization, along with the above factors. Efforts are done increasingly towards providing supplements which can maximize the human potential and improve sports performance, and hence supplements have been integral part of athlete's diet since last two decades. One must however, remember that sports supplements can't replace a balanced diet, proper training, rest or sleep, and also may carry risk of accidental or unintentional doping.

INTRODUCTION:

Sports nutrition plays a major role in improving sports performance and hastening recovery after an exhaustive exercise session (1,2). There are various sports in which one needs to maintain a low-calorie diet to maintain a certain weight, like sports of specific weight categories like boxing and judo etc. Some may need to maintain a low body fat like in gymnastics and ballet, and the athlete may do excessive total calorie reduction or reduced dietary lipids.

Assessment of nutritional status can be done by various ways like anthropometric analysis, biochemical tests, clinical assessment and monitoring diet intake etc. Economic status is equally important to evaluate the nutritional condition of an athlete, specially at the peripheral side of the country. Some of the commonly used techniques may be body mass index (BMI), skinfold thickness measurement, bioelectrical impedance analysis (BIA), Dual energy x-ray absorptiometry (DEXA) and hydrostatic weighing to get percentage body fat and lean body mass etc. A thorough search for articles on online databases like google scholar and pubmed was done with the key terms like "sports nutrients", "sports nutrition", "sports performance" and "recovery" and the only crux of the information was incorporated in this article regarding sports nutrition in sports performance and recovery.

BASICS ON SPORTS NUTRIENTS:

More than forty-five essential

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nutrients have been described which are important for energy requirement of body, normal body function and health, and which include carbohydrate, protein, fat, vitamins and minerals etc. They main not only provide energy but also help in growth, repair and recovery. The energy requirement varies in different sports and as per their level of activity and athlete's weight. For example, a gymnast may only require 3.5 kcal/min of energy as compared to 10.3 kcal/min in a judo player of 50 kg body weight (3–6). And the main source of energy is carbohydrate (65-70%), fat (25-30%) and proteins (10-15%). As per the sports and on the basis of daily physical activity, the requirements of carbohydrate may vary from 2g/Kg/day for sedentary persons to 10-12g/Kg/day in athletes of endurance events like triathlon ironman. Similarly, the daily proteins requirement may vary from 1g/kg/day to 2g/kg/day for sedentary person to ultra-endurance athlete respectively (4,6). The micronutrients like minerals and vitamins don't provide energy directly, but they are needed for maintenance of normal body functions like cell signaling pathways and oxidative processes at subcellular level. Proper hydration is also very important along with all the macronutrients and micronutrients required for normal body metabolic processes. In fact, our body is made up of 60-70% of water and water acts as a universal solvent, lubricant and medium for transport. The body's demand of macronutrients and micronutrients may get increased in athletes as compared to normal people due to increased physical activity due to sports activity and training. Normally it is recommended to average three days of dietary records for assessment of proper nutrition for normal people but average of seven days of dietary records are required for athlete's as per National Institute of Nutrition (NIN).

BASICS ON SPORTS SUPPLEMENTS:

Sports supplements are routinely given to athletes in all the sports discipline and it may play a role in enhancing sports performance and recovery. Some of the commonly used supplements may include whey or casein protein, BCAA (branch chain amino acids), multivitamin and multimineral, creatine, glutamine etc. The role of sport supplements, however, is controversial in sports performance, and hence balanced diet is always preferred. Sport supplements cannot make up for the lack of training, poor nutrition, inadequate rest etc. Although many studies are also there, which support the use of sports supplements. De Silva et al., in 2010 has done a study on 113 national level athletes in Sri Lanka and saw that after giving sport nutrients & supplements, sports performance has increased by 79.2% (7). Similarly, Aljaloud and Ibrahim, 2013 did a study on professional footballers in Saudi Arabia in 2013 and seen an improvement in sports performance by 43.8% after supplementation of sport nutrients in the diet (8).

CARBOHYDRATE LOADING & OTHER NUTRITIONAL INTERVENTION:

Historically carbohydrate loading and similar dietary intervention has been used with an aim to enhance performance. Although the evidence is weak or conflicting currently, especially in case of carbohydrate loading.

Many protocols have been described to improve sports performance via different diet interventions. One of the diet protocols described is divided into depletion phase and loading phase. Depletion phase consist of high intensity training and low carbohydrate intake for three days followed by three days of low intensity training and high carbohydrate intake in loading phase(9,10).It is also called as "train low, compete high" approach. The aim behind training with low carbohydrate intake is to promote cellular adaptations such as enhanced activation of cell signaling pathways, increased mitochondrial activity and increase lipid oxidation rates. There are other depletion protocols are also mentioned i.e., training after an overnight fast, training twice per day and restricted carbohydrate consumption during recovery(11).

Carbohydrate loading is done before a long duration exercise (>90min) to fulfill energy requirement, maintain glycogen stores and avoid hypoglycemia throughout the exercise session. It also plays a role in delaying onset of fatigue and improving sports performance(12). It may consist of intake of carbohydrate (200-300g, low glycemic index) 3-4 hours before exercise and consumption of carbohydrate (50-100g, high glycemic index) 01 hour before exercise.

Carbohydrate mouth rinse has also been shown to improve performance (4). It has been suggested that the oral receptors stimulate central nervous system to improve motor output during exercise and ultimately contributed to performance improvement in exercises lasting more than an hour (13).

Proper hydration is equally important in improving performance apart from all the macro and micronutrients. And it has been seen that loss of >2% of body mass can lead dip in performance (14). Isotonic fluids may be taken which contains carbohydrate (5-7g/100ml fluid) for an exercise session lasting more than 60 minutes to prevent dehydration and hypoglycemia specifically in hot and humid conditions. And there are many ways to avoid dehydration in athletes by looking at warning sign like dark yellow colored urine, excessive thirst and weight loss. To encourage hydration support staff should remind athletes to take water, keep plenty of water bottles on the field and encourage them by taking it yourselves at regular interval. But one should also be aware of excessive intake, specially of hypotonic solution, and hence risk of dilutional hyponatremia, specially in ultramarathon etc.

ROLE IN RECOVERY:

Sports nutrients plays a major role in enhancing recovery and delaying fatigue for example carbohydrate intake along with protein post exercise help in replenishing glycogen stores in muscle and repairing wear and tear in muscles. The glycogen synthesis is maximum at first hour post exercise and complete replenishment of glycogen stores can take up to 20 hours after an exhaustive exercise session.

DOPING POSSIBILITY:

There is always possibility of unintended and accidental doping with any kind of supplements which a player is taking. Many a times banned substances and chemicals may be there which may not be mentioned on the label. Hence, only supplements approved & certified by competent authority may be taken by the athletes, and proper recording should be done regarding everything which is consumed by an athlete. Athlete should never consume any over-the-counter supplements or any medicine without consulting their sports medicine doctors, or nutritionists.

CONCLUSION: -

On the basis of various studies and evidences, it is suggested that sports nutrition and supplements play a major role in enhancing sports performance and recovery. But it cannot negate the fact that the deficiencies present in diet, sleep, training and recovery cannot be overcome by providing supplements in diet alone. To improve sports performance athletes, coaches, nutritionist, sports medicine specialist and other supports staff should implement & follow a holistic approach to improve performance and overall health by providing a balanced diet, proper rest for recovery, recognizing signs of overtraining early and motivate athletes to maintain proper hydration throughout the training period as shown in figure 1. One need to be careful regarding supplement intake, as there is always possibility of accidental or unintentional doping.

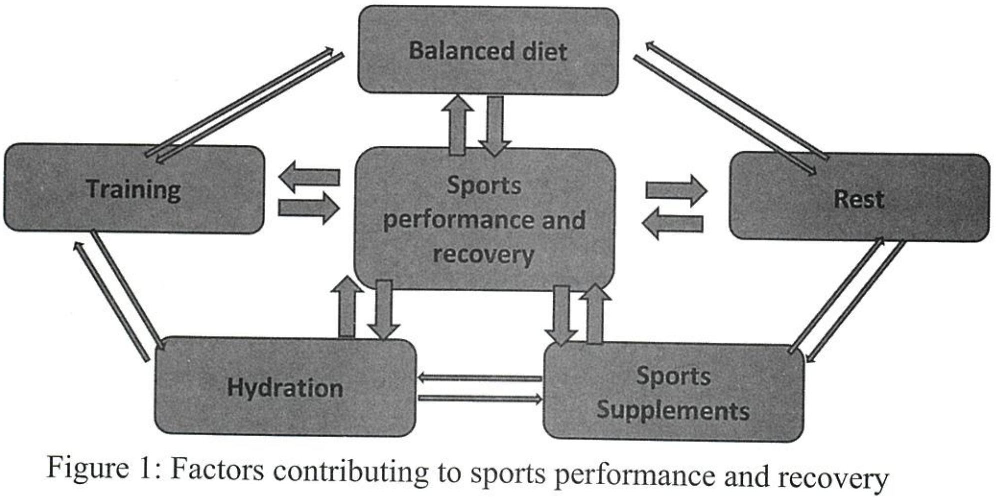

Figure 1: Factors contributing to sports performance and recovery

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